
Build the Geometry In, Don’t Hope It Emerges: A Reflection-Symmetry Audit of a Skeleton World Model

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Abstract

1 Does a self-supervised skeleton world model learn that the human body is bilat-
2 erally symmetric? We make the question exact by treating left–right reflection
3 as a group action, audit a frozen S-JEPA against it, and find a **robust informa-**
4 **tive null**—the symmetry does not emerge—that we then repair **by construction**.
5 The design principle: *build the geometry in; don’t hope it emerges*. All results
6 are transductive (internal validity only); the source video is the independent unit;
7 dataset folder labels are annotations, not diagnoses.

8 1 Bilateral symmetry as a $\mathbb{Z}/2$ group action

9 Reflecting a skeleton across the sagittal plane and swapping left/right landmarks yields a valid skele-
10 ton, so the mirror M (negate x ; swap all sixteen bilateral landmark pairs) satisfies $M^2 = I$ and
11 generates $G = \{I, M\} \cong \mathbb{Z}/2$. Define a **signed laterality** target $y(x) = \sum_k (\ell_k - r_k)$ over the
12 six bilateral pairs, where ℓ_k, r_k are per-joint temporal motion magnitudes. By construction it is
13 antisymmetric:

$$y(Mx) = -y(x), \quad \text{i.e.} \quad y(T_g x) = \rho(g) y(x), \quad \rho(M) = -1.$$

14 A world model “encodes bilateral symmetry” iff a read-out of its features reproduces this
15 antisymmetry—a **mirror slope of -1** —while remaining decodable. This is a *geometry-aware* probe
16 of an *articulated, deformable* body whose target is a *proprioceptive* quantity (which side moves
17 more)—the workshop’s themes made concrete (Fig. 1).

18 2 An audit protocol with ceiling, floor, and controls

19 We probe a frozen S-JEPA (33-joint BlazePose, masked latent infilling, curriculum-trained; check-
20 point fingerprint 7d13841a . . .) on five lanes: **A** learned per-pair token statistics; **B** raw-coordinate
21 ceiling ($R^2 \approx 1$); **C** untrained-encoder floor; **D** side-blind pooled control; **E** landmark-missingness
22 control. Estimation uses source-video-disjoint GroupKFold with an SVD-solver ridge and **repeated**
23 shuffled CV over the *fixed* cohort (10 reshuffles, mean $\pm$ a t -based across-reshuffle *stability* interval,
24 $\text{df} = 9$); the interval reflects partition sensitivity, not population sampling, and overlap is a stability
25 heuristic, not a significance test [Bengio and Grandvalet, 2004]. Pre-registered gates: A beats C by
26 ≥ 0.05 ; A reaches $\geq 80\%$ of B; sign correct on $\geq 75\%$ of held-out sources. The primary cohort is
27 the **626** sequences / **93** videos the encoder trained on (transductive); a 642 superset is a robustness
28 view.

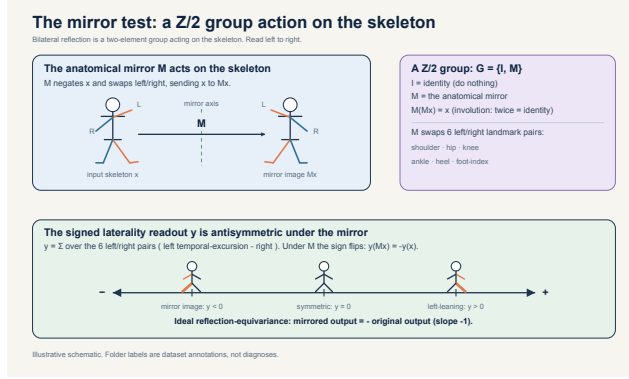


Figure 1: Bilateral reflection as a $\mathbb{Z}/2$ group action: the signed laterality target is antisymmetric by construction, so “encoding the symmetry” means a decoded mirror slope of exactly -1 .

Table 1: E1 laterality probe (626 primary).

Lane	Repeated-CV R^2 (stability interval)
A — learned	0.198 [0.175, 0.221]
C — untrained floor	0.245 [0.214, 0.275]
D — pooled	0.101 [0.089, 0.114]
E — missingness	0.202 [0.173, 0.231]
B — raw ceiling (single-partition sanity check)	≈ 1.000

3 E1: the symmetry does not emerge

Mirror slope -0.70 (no flip); sign consistency 0.58 . **All three gates fail**—under repeated CV the learned feature sits below the untrained floor in mean, with overlapping stability intervals (a partition-stability heuristic, not a significance test). The single-partition ordering $A > C$ reverses under repetition; an alpha sweep (score rising monotonically from -2.04 to $+0.27$ only under heavy regularization) is consistent with a weak, collinear feature. The missingness control E (0.202) has a mean close to A (0.198), flagging possible visibility confounding and corroborating the null. **A robust informative null.**

4 E2/E3: recover the geometry by construction

Frame-averaged read-out (E2). With $\Phi(x) = A(x) - A(Mx)$, a linear read-out on Φ has mirror slope **exactly -1 for any nonzero weights**. It reaches $R^2 = 0.273$ $[0.253, 0.294]$, above the free read-out 0.198 $[0.175, 0.221]$ in repeated-CV mean (disjoint stability intervals—a partition-stability heuristic, not a formal test). The controls are consistent with the antisymmetric *constraint* as the source of the gain (they do not establish a causal decomposition): the symmetric part Ψ scores 0.015 (slope $+1$, single-partition); an unconstrained $\geq$ -capacity MLP that sees both mirror passes but is untied collapses to 0.047 (slope -0.43 , single-partition); the exactly-tied nonlinear head adds nothing over linear Φ . The benefit over *learning* is only suggestive, though: frame-averaging the *untrained* encoder already reaches 0.219 $[0.175, 0.264]$ (overlapping interval). Point-estimate, Φ exceeds free A by 0.075 ; the architecture and learned-encoder contrasts ($\Phi_{\text{floor}} - A$, $\Phi - \Phi_{\text{floor}}$) are 0.021 and 0.054 with overlapping intervals—so the *learning* benefit is suggestive, not decisive.

Frame-averaged encoder (E3.1). Averaging the encoder, $E'(x) = \frac{1}{2}(E(x) + \sigma \cdot E(Mx))$, is **exactly token-equivariant with zero retraining** (equivariance error 0.0); the feature splits exactly into antisymmetric $(\ell - r)$ and symmetric $(\ell + r)$ blocks. The antisymmetric block *corresponds to* E2’s Φ —it is one half of Φ ’s $\ell - r$ sub-block (up to standardization), while Φ additionally antisymmetrizes the $\ell + r$ block.

Augmentation does not induce it (E3.2, single-seed). Retraining Stage-0 with reflection augmentation ($p=0.5$ vs 0.0 , one switch) changes the loss but not the geometry: the augmented and untrained axes have similar repeated-CV means (0.062 $[0.011, 0.114]$ vs 0.078 $[0.013, 0.144]$); over-

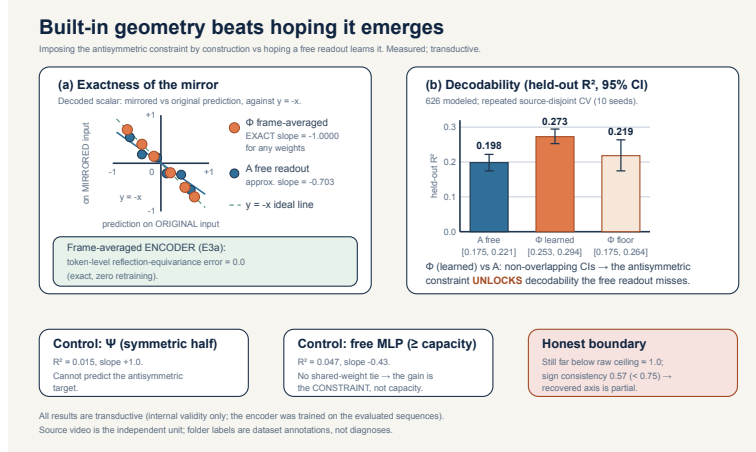


Figure 2: Built-in beats emergent, at both the read-out and the encoder. Emergent (E1/E3.2): at or below floor, slope not -1 . Built-in (E2/E3.1): frame averaging recovers exact antisymmetry.

lapping stability intervals), so the negative rests on the mirror slope, which moves no closer to -1 (the *untrained* encoder is closest, -0.82 ; the augmented arm furthest, -0.51 —in this single-seed arm, slope proximity is not evidence of learned equivariance, consistent with sensitivity to feature geometry). A single-seed negative.

5 Synthesis and takeaway

The four experiments form a 2×2 grid (Fig. 2)—symmetry in the read-out or the encoder, hoped-for or built-in. *Read-out*: emergent E1 has slope -0.70 with learned $\approx$ floor (intervals overlap), while built-in E2 has slope -1.0000 and $0.273 > 0.198$. *Encoder*: emergent E3.2 has slope not toward -1 and axis $\leq$ floor, while built-in E3.1 has token error 0.0 (exact). In the audited system (single checkpoint; single-seed augmentation arm), standard and reflection-augmented self-supervision leave bilateral symmetry un-learned; frame averaging recovers it to machine precision at the read-out or encoder. The constructive claim is general; the negative is scoped to what we tested. For geometry-aware world models of articulated bodies, symmetry is cheap to *impose* and—at least here—not reliably obtained by *hoping* it emerges: **build the geometry in; don’t hope it emerges**.

Ethics and limitations

Results are transductive (internal validity only; no generalization claim). The dataset (GAVD [Ranjan et al., 2025]) provides annotations and public YouTube URLs, not raw video; users retrieve media independently under YouTube’s terms; we use derived pose sequences, infer no identity, and redistribute no raw or identity-bearing frames. **No institutional ethics determination or completed data-use review is yet on record; both must be resolved before submission.** Condition folder labels are dataset annotations, not diagnoses; the independent unit is the source video, not the individual; no clinical claim is made.

References

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